



# **PREDICTIVE EQUIPMENT MAINTENANCE AGENT**

**TEAM DATA MINERS**



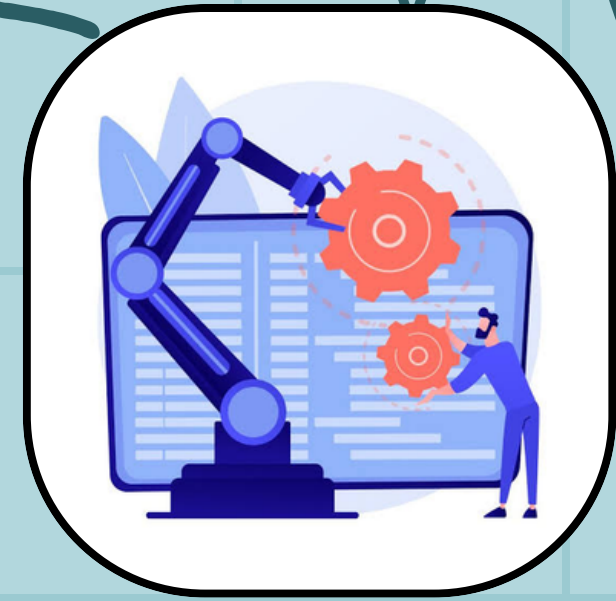
# INTRODUCTION

- Industrial machines continuously generate sensor data such as temperature, vibration, pressure, RPM, and power consumption.
- This data contains important information about the operating condition and health of equipment.
- Our project uses data analysis and machine learning to identify abnormal machine behavior and predict potential equipment failures.
- The system converts raw sensor data into failure risk information and maintenance recommendations.
- The goal is to shift maintenance from a reactive approach to a predictive approach



# Problem statement

- Unexpected equipment failures can cause production downtime, repair costs, productivity loss, and safety risks.
- Traditional reactive maintenance responds only after a machine fails.
- Preventive maintenance follows fixed schedules and may not reflect the machine's actual condition.
- Large volumes of sensor data are difficult to analyze manually and continuously.
- There is a need for a system that can automatically detect abnormal conditions and identify potential failure risks.



# Proposed solution

Sensor Dataset



Data Validation & Preprocessing



Time-Series Analysis



Sensor Trend Visualization



Abnormal Pattern Detection



Failure Prediction



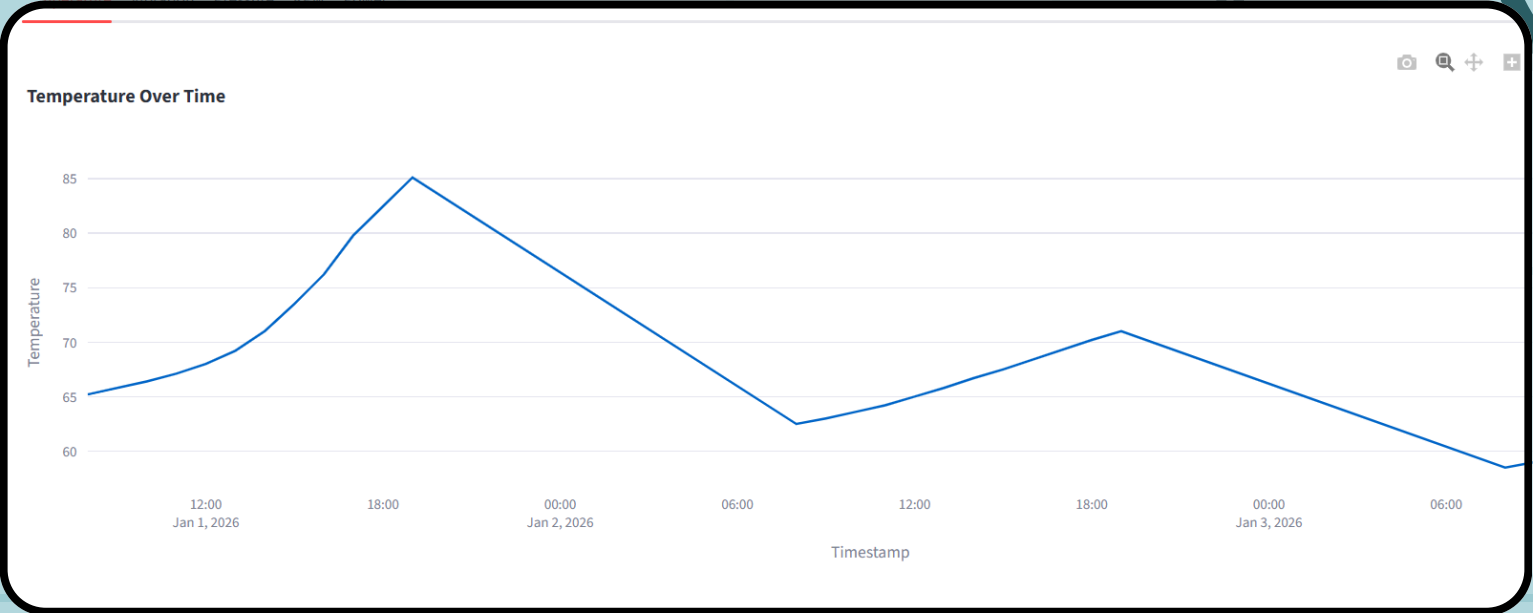
Failure Probability



Risk Classification



Maintenance Alert



**Predictive Equipment Maintenance**

Welcome to the Predictive Equipment Maintenance portal. Upload equipment sensor data in CSV format to get started.

Upload CSV file

predict...sample.csv 1.3KB

File uploaded successfully: predictive\_maintenance\_sample.csv

**Data Quality Summary**

|            |                    |                        |               |
|------------|--------------------|------------------------|---------------|
| Total Rows | Duplicates Removed | Missing Values Imputed | Total Columns |
| 30         | 0                  | 0                      | 6             |

Time Range: 2026-01-01 08:00:00 to 2026-01-03 13:00:00

**Cleaned Data Preview (First 5 Rows)**

|   | timestamp           | temperature | vibration | pressure | rpm  | power |
|---|---------------------|-------------|-----------|----------|------|-------|
| 0 | 2026-01-01 08:00:00 | 65.2        | 5.2       | 2.1      | 1450 | 18.5  |
| 1 | 2026-01-01 09:00:00 | 65.8        | 5.2       | 2.2      | 1455 | 18.7  |
| 2 | 2026-01-01 10:00:00 | 66.4        | 5.1       | 2.3      | 1460 | 18.9  |
| 3 | 2026-01-01 11:00:00 | 67.1        | 5.1       | 2.5      | 1465 | 19.1  |

**Predictive Equipment Maintenance**

Welcome to the Predictive Equipment Maintenance portal. Upload equipment sensor data in CSV format to get started.

Upload CSV file

Upload 200MB per file • CSV

# HOW IT WORKS



- The user provides equipment sensor data.
  - The system cleans and analyzes the data over time.
  - Sensor trends and statistical characteristics are evaluated.
  - Isolation Forest identifies unusual operating patterns.
  - Random Forest predicts potential failure using historical failure data.
  - The system generates a failure probability and risk level.
  - Finally, it provides a maintenance recommendation based on the identified risk.
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# IMPORTANCE AND CONCLUSION

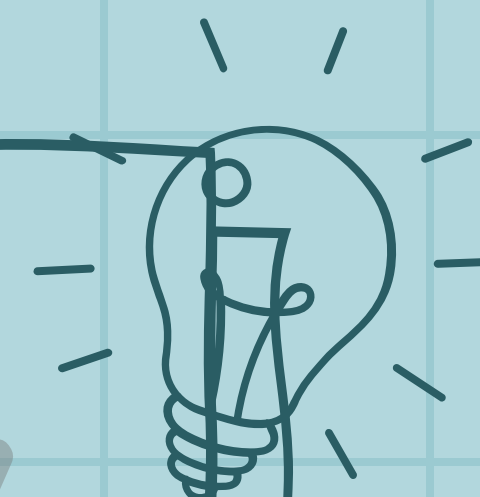


## Why Is It Important?

- Helps identify potential equipment problems before complete failure.
- Supports data-driven maintenance decisions
- Can help reduce unexpected downtime and unnecessary emergency maintenance.
- Converts complex sensor data into simple and actionable information
- Provides a foundation for future real-time industrial predictive maintenance.

## Conclusion

- The Predictive Equipment Maintenance Agent transforms raw equipment sensor data into actionable maintenance intelligence. By combining time-series analysis, anomaly detection, machine learning, and risk-based alerts, the system supports the transition from reactive maintenance to predictive maintenance
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**THANK  
YOU**

